

Deepak Mallampati

(330) 554-0770 | deepakmallampati00@gmail.com | linkedin.com/in/deepak-mallampati |

Kent, OH - Open to relocation to Paris (visa sponsorship required)

PROFESSIONAL SUMMARY

Full-stack engineer with AI/LLM specialization and production systems delivery experience (2.5+ years). Master's degree (Kent State University). Built and shipped end-to-end systems across healthcare AI (FastAPI/Docker production services, CI/CD pipelines, compliance-grade data governance) and independently-developed full-stack applications (React/TypeScript, multi-provider AI integration). Track record of measurable outcomes: ~30% fewer post-deployment defects, ~35% CI/CD release cycle improvement, ~25% coordination overhead reduction. Bringing systems integration discipline and AI capability to Mistral AI's Enterprise Agents team.

CORE COMPETENCIES

Python / FastAPI / Flask • TypeScript / React • CI/CD (Jenkins) • Docker • SQL (PostgreSQL, T-SQL, SQLite) • LLM / Agentic Workflows • RESTful API Design • Observability / Grafana • System Architecture • Stakeholder Communication • Greenfield System Design • Compliance-Sensitive Environments

PROFESSIONAL EXPERIENCE

Progress Solutions Inc.

Jul 2025 - Present

Jr. AI/ML Engineer Trainee - Healthcare AI, USA

- Packaged ML/LLM inference as **FastAPI/Flask RESTful services**, containerized with Docker, with structured logging and load simulation - reduced post-deployment defects ~**30%**.
- Built **agentic LLM workflows** for multi-step healthcare queries - tool discipline, grounding rules, structured outputs; reduced hallucination >30%, improved agent stability ~25%.
- Maintained **HIPAA-conscious data governance**: de-identification, data lineage, audit trails, and stakeholder-facing system documentation.
- Built preprocessing pipelines for EHR extracts (Pandas, NumPy) - raised dataset reliability above **98%**.

Energy Solutions International (Emerson)

Jun 2022 - Apr 2023

Database & DevOps Performance Engineer (Intern) - Hyderabad, India

- Built **Jenkins CI/CD pipelines** for SQL schema updates and stored procedure deployments; version-controlled releases with rollback checkpoints - reduced deployment errors >**30%**, improved release cycles ~**35-40%**.
- Designed **Grafana dashboards** (SQL Server DMVs) for performance monitoring - cut incident recurrence ~**25%** and improved root-cause resolution ~**30%**.
- Optimized T-SQL stored procedures for Synthesis Order-to-Cash (contracts, nominations, invoicing) - reduced query execution time ~**20%** and data retrieval latency ~**25%**.
- Supported RBAC implementation and audit logging in a compliance-sensitive oil & gas environment.

Kent State University

Aug 2023 - May 2025

Student Manager - Operations

- Led coordination for **150+ students**; formalized onboarding cutting ramp-up time ~**20%**; structured scheduling reduced coordination delays ~**30%**.
- Managed event planning and vendor comparison; achieved **10-15% cost efficiency** in select initiatives.

SELECTED PROJECTS

Manga Lens - Chrome Extension for Real-Time AI Translation

shipped - Chrome Web Store

Full-stack browser extension - Manifest V3, TypeScript, React frontend, service workers. Integrates **four AI vision providers**. Multi-provider payload handling, viewport-slice screenshot pipeline, 7-day cache, narrowed host permissions, privacy policy. Built and shipped independently end-to-end.

TypeScript, React, Python, Manifest V3, multi-provider AI APIs

Dream Decoder - AI-Powered Multimodal Creative Platform

Full-stack FastAPI + React/TypeScript/Vite/Tailwind application with coordinated multimodal API integrations (Perplexity Sonar + Replicate image models). Structured prompt transformation layers improved contextual alignment ~**30%**.

FastAPI, React, TypeScript, Tailwind, multi-modal AI orchestration

Agentic Healthcare Claims Processing System

Multi-agent production pipeline with schema-validated JSON contracts between agents, RAG-grounded validation, and audit-ready reasoning traces. Greenfield system designed, built, and deployed end-to-end - demonstrates technical ownership and systems integration discipline.

LangChain, FastAPI, Docker, Python

E-Farming Digital Marketplace founded

Independently designed and shipped a PHP/MySQL/Bootstrap AgriTech marketplace. Onboarded **80-120 active users** in phase one. Full stack: cart, reviews, community blogging, seller/buyer accounts.

PHP, MySQL, Bootstrap, full-stack web development

EDUCATION

Master's Degree - Kent State University, Ohio, USA

Aug 2023 - May 2025

TECHNICAL SKILLS

Languages & Frameworks: Python, FastAPI, Flask, TypeScript, React, PHP, SQL (PostgreSQL, T-SQL, SQLite)

LLM & AI (nice-to-have coverage): LangChain, LlamaIndex, RAG, agentic workflows, structured outputs, prompt engineering

Infra & DevOps: Docker, Jenkins CI/CD, Grafana, RESTful APIs, cloud-ready deployment, observability/logging

Data: Pandas, NumPy, SQL optimization, T-SQL stored procedures, RBAC, audit logging

Work Authorization: F-1 OPT (US-authorized); requires visa sponsorship for France/EU